

Part-II Literature review and technical reading

• New and Existing knowledge -

- New knowledge must be interpreted within the context of existing knowledge.
- Existing knowledge forms the foundation that supports and validates new findings.
- The significance of new knowledge depends on what is already known and understood.
- The existing knowledge helps identify the problem gap - what is missing or known.
- Knowledge foundation is built through reading and surveying literature.
- Textbooks provide established, foundational knowledge.
- A strong knowledge base, supported by relevant literature, makes new research credible and hard to challenge.

• Literature review -

- Shows how current research connects with past work.
- Explains why and how earlier studies were done.
- Builds a strong base for new knowledge.
- Focus on concepts, not just authors or papers.
- Should - Tell what is already known, point out key ideas and factors, mention existing methods or findings, note any gaps, conflicts or missing points, and give reasons why more research is needed.

A good literature survey is a two-step process -

- (1) Identify the major topics or concepts relevant to subject under consideration.
- (2) Place the citation of the relevant source.

• Analysis and synthesis of prior art -

A researcher should analyze the relevant information by undertaking following steps -

- (i) Understanding the hypothesis.
- (ii) Understanding the models and the experimental conditions used.
- (iii) Making connections.
- (iv) Comparing and contrasting the various information.
- (v) Finding out the strong points and loopholes.

• Bibliographic Databases -

- They are the abstracting and indexing services for research literature.
- They provide citations and often abstracts of scholarly articles for easy search.
- Enable researchers to search across large collections of academic work.
- Using multiple databases prevents reliance on just one and avoid its limitations.
- Researchers must identify databases relevant to their topic.
- Commonly used databases are those popular among engineering researchers - Web of Science, Google, and Google Scholar.

• Conceptualizing research -

A research objective should be original, significant, and knowledge oriented.

It must be practical and achievable using suitable methods and tools.

Formulating a strong objective requires thorough understanding of existing literature.

Research should aim to generate new knowledge, not merely create or build something obvious.

- Critical and creative reading -
- Reading a research paper requires careful and analytical thinking.
- Critical reading focuses on identifying errors and strong ideas in the work.
- Creative reading involves finding new applications or extension of the study.
- It encourages original thinking and exploring possible improvements or practical challenges.

• Citations: Functions and attributes

- Citations credit original authors and help readers trace the source of information.
- Each source must be cited in-text and again in the reference list.

• Functions of citations:

Verification - ensures accuracy and prevents misleading information.

Acknowledge - Give due credit and enhances researcher reputation.

Documentation - Records scientific progress and historical development.

Improper citations reduce value, such as spurious, biased, self - or coercive citations.

• Impact of Title and Keywords on citations -

- Citation rate depends on journal quality, publication type, research area, and importance of work.
- Title, length, type, and chosen keywords also influence citation count.

• Title -

Title is a key attribute of any research paper.

Plays an important role in marketing and traceability of research.

A good title is clear, informative, and attracts readers' attention.

Title behavior depends on -

- Type of title
- Length of title
- Presence of specific markers

• Keywords -

Show the main ideas of a research paper.

They help search engines and journals find and group similar studies.

Good keywords make the papers easy to find for the right readers.

They ensure readers understand what the paper is about.

Refer knowledge flow through citation, acknowledgements and attributions from ppt.